

Lecture 33

STAT 109: Introductory Biostatistics — Chi-square test of independence

Lecture 33

Chi-square test of independence for two categorical variables

When both the explanatory and response variables are categorical, we summarize the data in a contingency table of counts (each cell = number of units with that combination of levels). The Pearson chi-square test of independence asks whether the data are compatible with independence in the population:

- H_0 : the two variables are independent (knowing one does not change the distribution of the other).
- H_a : the two variables are not independent (there is association).

In R, build a table of counts (with `table()` or `count() + pivot_wider()`), then run `chisq.test(tab)`. Inspect `chisq.test(tab)$expected` for the **expected counts** under independence.

Conditions.

- Independent observations (e.g. one row per person, or clearly independent trials as designed).
- For the usual Pearson p -value, all expected cell counts should be at least about 5. If not, use `simulate.p.value = TRUE` in `chisq.test`, collect more data, or collapse sparse categories.

Plots. Stacked bar charts (counts or 100% filled)

Example 1 — RCT: three vaccine arms vs flu (yes/no)

Story. Healthy adults were **randomly assigned** to one of **three** study arms (**Placebo**, **Low-dose vaccine**, **High-dose vaccine**). Over winter follow-up, whether each person had **influenza (yes/no)** was recorded.

Research Question: the chi-square test addresses whether **flu status** is **independent** of **arm** in the population of people like those enrolled (equivalently, whether **flu rates** differ across arms).

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

set.seed(3301)
n_per <- 55
p_flu <- c(Placebo = 0.22, `Low dose` = 0.12, `High dose` = 0.08)
flu_rct <- tibble(
  arm = factor(
    sample(rep(names(p_flu), each = n_per)),
    levels = c("Placebo", "Low dose", "High dose")
  )
) |>
mutate(
  flu = factor(
    rbinom(n(), 1, p_flu[as.character(arm)]),
    levels = c(0, 1),
  )
)
```

```

      labels = c("No", "Yes")
    )
  )

```

Table of counts

```

flu_rct |>
  count(arm, flu)

```

```

# A tibble: 6 x 3
  arm      flu      n
  <fct>    <fct> <int>
1 Placebo  No       39
2 Placebo  Yes      16
3 Low dose No      44
4 Low dose Yes     11
5 High dose No     49
6 High dose Yes      6

```

Counts in wide form (rows = arm, columns = flu)

```

wide_counts <- flu_rct |>
  count(arm, flu) |>
  pivot_wider(names_from = flu, values_from = n, values_fill = 0)
wide_counts

```

```

# A tibble: 3 x 3
  arm      No  Yes
  <fct>    <int> <int>
1 Placebo    39   16
2 Low dose   44   11
3 High dose  49    6

```

Row proportions (flu rate within each arm)

```

flu_rct |>
  count(arm, flu) |>
  group_by(arm) |>
  mutate(prop_within_arm = n / sum(n)) |>
  ungroup()

```

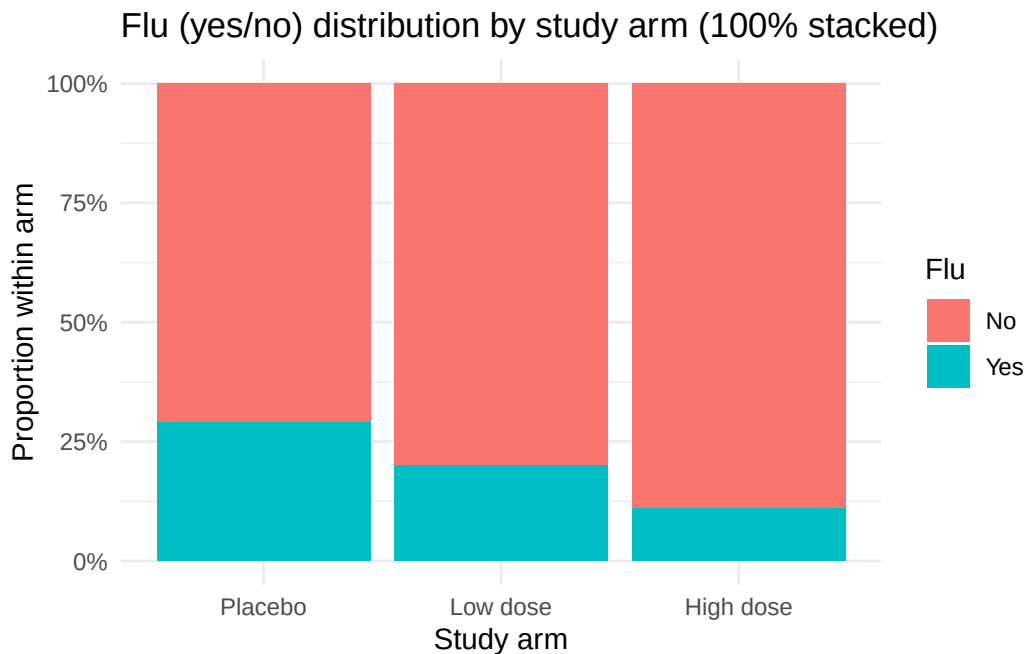
```

# A tibble: 6 x 4
  arm      flu      n prop_within_arm
  <fct>    <fct> <int>          <dbl>
1 Placebo  No      39          0.709
2 Placebo  Yes     16          0.291
3 Low dose No     44           0.8
4 Low dose Yes    11           0.2
5 High dose No     49          0.891
6 High dose Yes      6          0.109

```

100% stacked bar chart (proportion with/without flu, by arm)

```
flu_rct |>
  count(arm, flu) |>
  ggplot(aes(x = arm, y = n, fill = flu)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Flu (yes/no) distribution by study arm (100% stacked)",
    x = "Study arm",
    y = "Proportion within arm",
    fill = "Flu"
  ) +
  theme_minimal()
```



Chi-square test of independence

```
tab_flu <- table(Arm = flu_rct$arm, Flu = flu_rct$flu)
tab_flu
```

Arm	Flu	
	No	Yes
Placebo	39	16
Low dose	44	11
High dose	49	6

```
chisq.test(tab_flu)$expected
```

Arm	Flu	
	No	Yes
Placebo	44	11
Low dose	44	11

High dose 44 11

```
chisq.test(tab_flu)
```

Pearson's Chi-squared test

data: tab_flu

X-squared = 5.6818, df = 2, p-value = 0.05837

Interpretation. If expected counts are fine and the p -value is small, the sample is **not** consistent with **independence** between arm and flu; describe **which arms** have higher **observed** flu fractions using the table or plot.

Example 2

Story. Shoppers at a campus store were randomly shown one of three package designs (D1, D2, D3) for the same snack; whether they purchased (yes/no) was recorded.

```
set.seed(3302)
n_per <- 48
p_buy <- c(D1 = 0.35, D2 = 0.48, D3 = 0.40)
pack_rct <- tibble(
  design = factor(
    sample(rep(names(p_buy), each = n_per)),
    levels = c("D1", "D2", "D3")
  )
) |>
mutate(
  purchase = factor(
    rbinom(n(), 1, p_buy[as.character(design)]),
    levels = c(0, 1),
    labels = c("No", "Yes")
  )
)
```

Counts and column proportions (purchase rate by design)

```
pack_rct |>
count(design, purchase)
```

```
# A tibble: 6 x 3
  design purchase     n
  <fct>   <fct>   <int>
1 D1     No       33
2 D1     Yes      15
3 D2     No       28
4 D2     Yes      20
5 D3     No       31
6 D3     Yes      17
```

```
pack_rct |>
count(design, purchase) |>
```

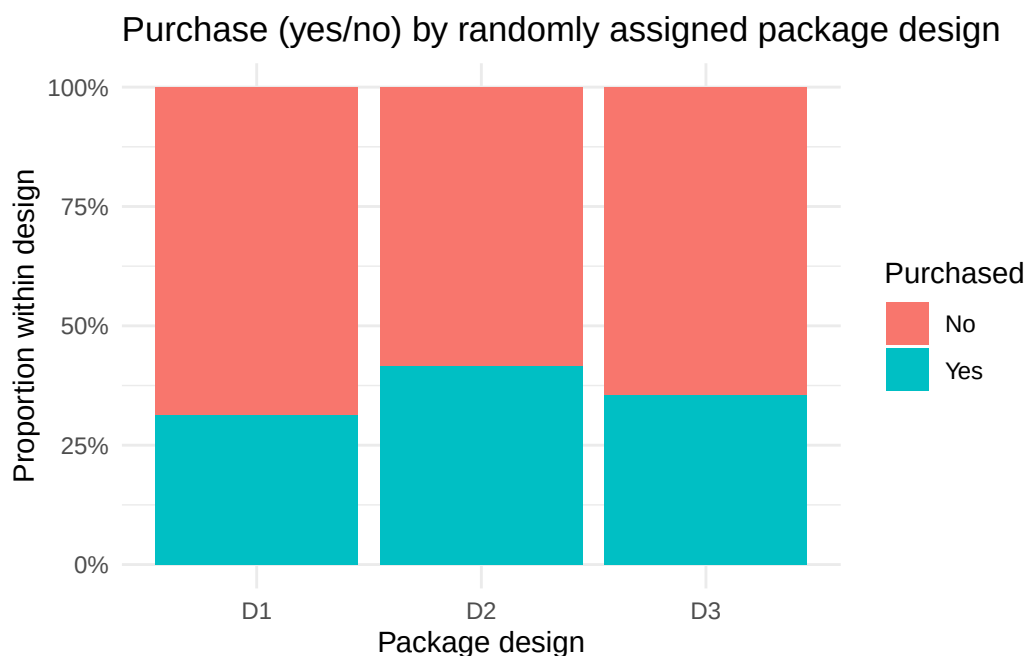
```
group_by(design) |>
mutate(prop_within_design = n / sum(n)) |>
ungroup()
```

A tibble: 6 x 4

	design	purchase	n	prop_within_design
	<fct>	<fct>	<int>	<dbl>
1	D1	No	33	0.688
2	D1	Yes	15	0.312
3	D2	No	28	0.583
4	D2	Yes	20	0.417
5	D3	No	31	0.646
6	D3	Yes	17	0.354

100% stacked bar chart

```
pack_rct |>
count(design, purchase) |>
ggplot(aes(x = design, y = n, fill = purchase)) +
geom_col(position = "fill") +
scale_y_continuous(labels = scales::percent_format()) +
labs(
  title = "Purchase (yes/no) by randomly assigned package design",
  x = "Package design",
  y = "Proportion within design",
  fill = "Purchased"
) +
theme_minimal()
```



Chi-square test of independence

```
tab_pack <- table(Design = pack_rct$design, Purchase = pack_rct$purchase)
tab_pack
```

	Purchase	
Design	No	Yes
D1	33	15
D2	28	20
D3	31	17

```
chisq.test(tab_pack)$expected
```

	Purchase	
Design	No	Yes
D1	30.66667	17.33333
D2	30.66667	17.33333
D3	30.66667	17.33333

```
chisq.test(tab_pack)
```

Pearson's Chi-squared test

data: tab_pack

X-squared = 1.1438, df = 2, p-value = 0.5644

Conclusion. Reject that the rates are all equal if the p-value is small. Follow up with descriptive proportions and plots to say how purchase differs across designs—especially important when reporting to a client or writing a methods paragraph.